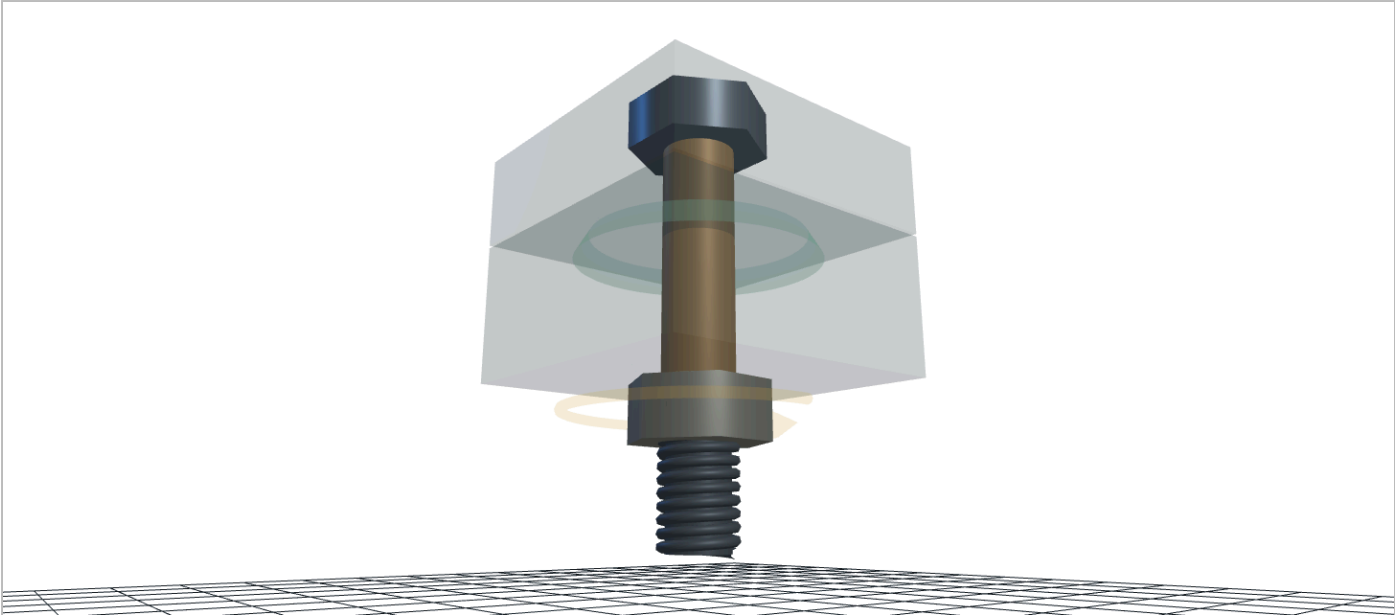


Bolted Joint — design calculation

M6 (Ø6 mm · As 20.1 mm²) · class 8.8 (medium-carbon, Q&T) · Dry steel, plain (K ≈ 0.20) · no washers
Plate 1 Aluminum 6061-T6 × 8.00 mm · plate 2 Mild steel (S235) × 12.00 mm · grip 20.00 mm · preload target 65% of proof – conservative (default)
T = 6.00 N·m → Fi 5.00 kN · external load P = 500 N · MechCalc – design check, not a substitute for full analysis

n = 1.80 · reduced 322 MPa vs proof 580 MPa while torquing · clamp left 4.63 kN at P · separation n 13.49



The joint as modelled at T = 6.00 N·m: the cone is the clamp load spreading through the plates, shaded by local pressure; bolt and plates are coloured by how close each is to its own limit. Deflections exaggerated x40 so the motion is visible.

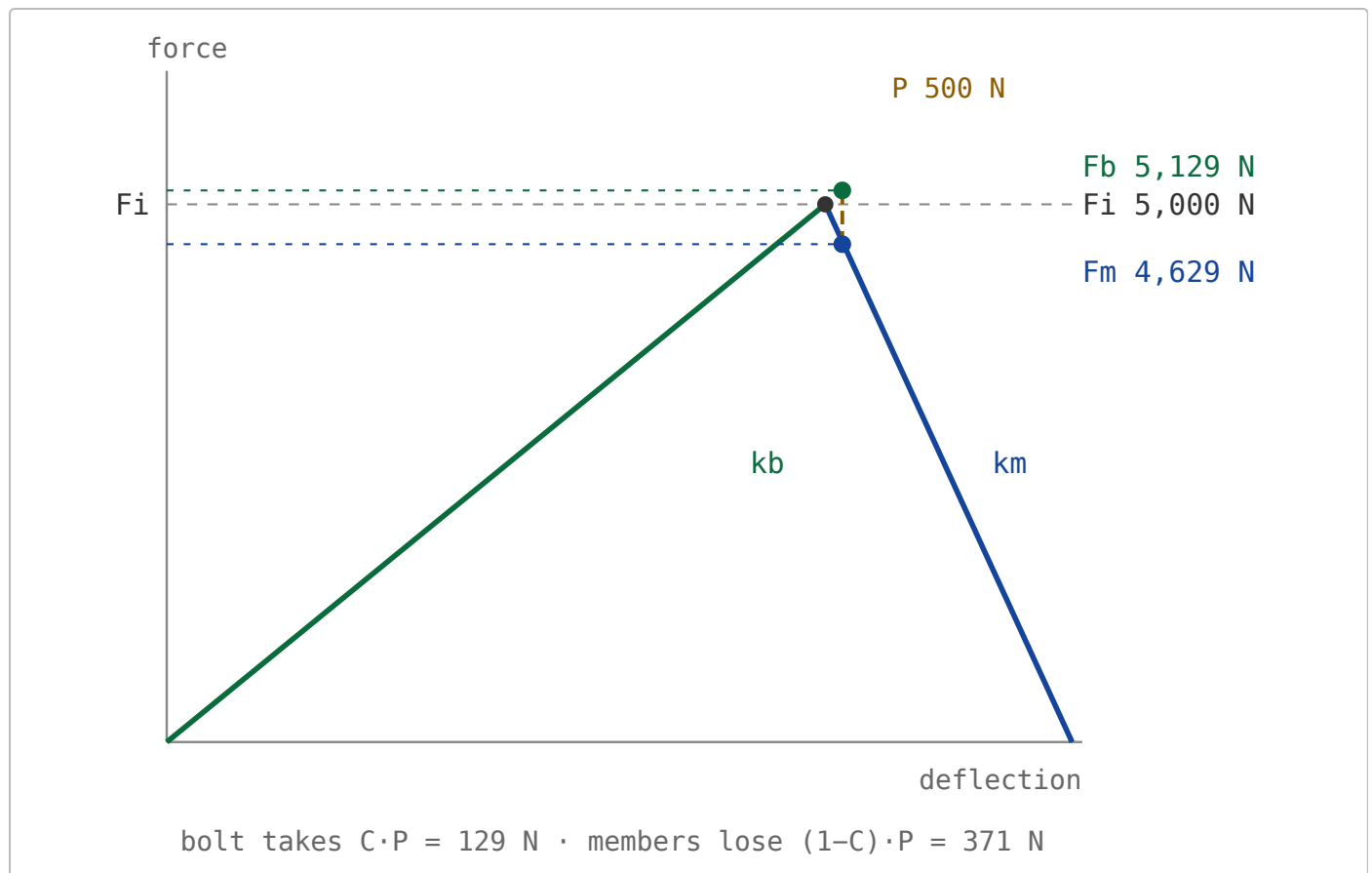
Joint at a glance

Preload Fi from T	5.00 kN
Reduced σred (vM) while torquing	322 MPa · n 1.80 vs proof
Working σ after torsion relaxes	255 MPa · n 2.51 vs yield
Stiffness kb / km · load share C	201 / 576 kN/mm · C 0.259
Bolt force / clamp left at P	5.13 kN / 4.63 kN
Separation load Psep	6.74 kN · n 13.49
Bearing p under head/nut	161 MPa · n 1.86 / 3.04
Interface pressure · cone Ø	20 MPa · Ø18.2 mm
Bolt stretch / plates squash	24.9 / 8.7 µm
Recommended torque	9.09 N·m · limited by the bolt

Equations used

$F = T / (K \cdot d)$ – Preload from tightening torque – nut-factor form
 $\sigma = F / A_s$ – Direct tensile stress on the stress area A_s
 $\tau = 16 \cdot T_{th} / (\pi \cdot d_s^3)$, $T_{th} \approx 0.5 \cdot T$ – Torsion from thread friction while torquing
 $\sigma_{red} = \sqrt{\sigma^2 + 3\tau^2}$ – Reduced (von Mises) stress during tightening – VDI 2230
 $n = S_p / \sigma_{red}$ – Safety factor against proof strength
 $k_b = E \cdot A_s / L$ – Bolt as a tension spring over the grip
 $k_m = f(30^\circ \text{ cone frusta})$ – Member stiffness – Shigley pressure-cone stack
 $C = k_b / (k_b + k_m)$ – Stiffness ratio: bolt's share of external load
 $F_b = F_i + C \cdot P$, $F_m = F_i - (1 - C) \cdot P$ – Load sharing: bolt force & remaining clamp
 $P_{sep} = F_i / (1 - C)$ – External load at which the joint separates
 $p = F / [\pi(d_w^2 - d_h^2)/4]$ – Bearing pressure under head/nut vs plate limit pG
 $T_{rec} = K \cdot d \cdot \min(0.65 \cdot S_p \cdot A_s, 0.9 \cdot p_G \cdot A_{bear})$ – Recommended torque – bolt or plate, whichever gives out first

The joint diagram — who carries the load



A bolted joint is a pre-stretched bolt fighting pre-squashed plates. Tightening does not "hold the parts together" by pulling on them — it locks a large internal force into the assembly, and every external load after that is a change to how that locked-in force is *shared*. All the interesting behaviour, good and bad, comes out of the sharing.

What the numbers mean

Preload F_i is what the wrench installs. **C** is the bolt's share of anything that comes later. **F_m** is the clamp left at the interface — friction, sealing and stiffness all come from it, and when it reaches zero the joint has separated even though nothing has broken. **σ_{red}** is the bolt's worst moment, mid-wrench, with torsion still in the shank.

Tightening

The wrench torque drives the nut down the thread incline, converting twist into axial preload that clamps the plates. While the wrench is on, thread friction also twists the shank, so the tightening check has to combine tension and torsion into the *reduced stress* $\sigma_{red} = \sqrt{(\sigma^2 + 3\tau^2)}$. This von Mises form is the standard bolted-joint method (VDI 2230; Shigley arrives at the same numbers by capping preload at 75–90% of proof instead). That torsion largely dissipates once the wrench is released, which is why the calculator also reports the milder **working stress** — pure tension, including the bolt's share of the external load — checked against yield rather than proof.

The clamped sandwich

The joint is two springs in parallel: the bolt (stiffness k_b) stretched by F_i , and the plate stack (k_m) compressed by the same F_i . The clamp force does not flow uniformly through the plates — it spreads in $\sim 30^\circ$ pressure cones from under the head to mid-grip and back to the nut, which is what the frustum stiffness model captures. Each plate's material enters through its modulus: swap a steel plate for aluminium or POM and k_m drops, shifting $C = k_b / (k_b + k_m)$. Stiff plates (small C) are why preloaded joints survive fatigue — the bolt barely notices the load cycles. But the clamp erodes by $(1 - C) \cdot P$, and at P_{sep} the plates separate; after that the bolt takes everything, and the joint hammers itself apart.

What "external load P" means

P is the working load your product applies to the joint *after* it is assembled — the thing trying to pull the two plates apart along the bolt axis. Pressure lifting a cover, a belt tensioning a bracket, the weight of whatever hangs off the part, an impact. It is *not* the tightening force: preload comes from the torque, and P is what arrives later, in service. The interesting result is that the bolt does not feel all of P — it picks up only $C \cdot P$, typically 10–25% with metal plates, while the remaining $(1-C) \cdot P$ is subtracted from the clamp squeezing the plates. Set $P = 0$ to look at the tightened joint alone; raise it to find the load where the clamp reaches zero and the joint separates.

Soft materials and bearing

The head and nut press on small annular faces, and soft plate materials crush there long before the bolt is in danger — so the calculator checks that surface pressure against each material's permissible pressure p_G (VDI-style values). This is also why the **recommended torque follows the materials you clamp**, not just the bolt: the target preload is the lesser of 65% of the bolt's proof load and what the softer plate's bearing limit allows. With steel or aluminium plates the bolt governs and you get the familiar handbook number (9.00 N·m for M6 class 8.8, dry); put the same bolt through nylon or FR-4 and the recommendation drops several-fold, because the plate would crush first. That matches the separate torque tables plastics and PCB suppliers publish. Washers raise the limit by enlarging the bearing area — worth adding whenever a plate flags red. Embedding (surfaces flattening over time) also costs proportionally more preload in soft, short joints.

Reading the diagram above

The two lines are the springs. The **shallow** one is the bolt — it stretches a lot for little force. The **steep** one is the plate stack, which barely moves. They meet at the preload. An external load walks the joint point to the right: the bolt climbs its shallow line by $C \cdot P$ while the members fall down their steep one by $(1-C) \cdot P$. Here that is **129 N** onto the bolt against **371 N** off the clamp. Where the member line hits zero, the plates lift and the geometry stops applying — beyond that point the bolt is alone.

Reading the 3D view

The cone inside the plates *is* the clamp load, spreading at 30° from under the head to mid-grip and converging on the nut. It is shaded by the local pressure at each depth: intense at the two bearing faces where the force crosses a small annulus, washed out in the middle where the same force is carried by a much larger area. That is exactly why fasteners crush the surface long before they crush the core, and why a washer fixes it. Each depth is coloured against the limit of whichever plate it falls in, so in a mixed stack the soft half goes amber while the steel beside it stays cool under the identical force.

Where this model stops being right

- 1 • Torque control is the weak link, not the arithmetic.** Every preload figure here inherits K's $\pm 25\%$ scatter. The equations are exact; the input is not. Joints that matter use angle control past snug, or measure bolt stretch directly.
- 2 • Embedding and relaxation are not modelled.** Real surfaces flatten over the first hours and days, and short, soft joints lose proportionally more of their preload to it — a thin plastic stack can give back 10–20%. Re-torque after 24 h, or design with the loss allowed for.
- 3 • The load is assumed concentric and purely tensile.** An eccentric load bends the bolt and shifts the pressure cone off-axis; a shear load is carried by friction at the interface (or by a dowel), not by the bolt in the way this page describes. Neither is checked here.
- 4 • Fatigue, thread stripping and gaskets are out of scope.** The stiffness ratio C is the first half of a fatigue calculation, but the alternating stress and the thread-root concentration are not evaluated. Engagement length is assumed sufficient — in a tapped soft-metal or plastic hole it usually is not, and the threads strip before any number on this page is reached.
- 5 • The 3D view exaggerates the deflections.** Real stretch here is 24.9 μm and real squash is 8.7 μm — invisible at any honest scale, so the viewer multiplies both by 40 $\times$. The kinematics are true even so: the head only sinks as the plates compress, and the elongation appears below the nut where it really does.

In short: tighten until the bolt carries a healthy fraction of proof — checked with torsion included, von Mises — and make the plates as stiff as you can. Then external loads mostly relax the plates instead of working the bolt. Watch the two clamped

materials: they set how the load is shared, when the joint separates, and whether anything crushes under the head.

Worked calculation

GIVEN	VALUE
Thread M6 – $d \cdot A_s$	6.00 mm · 20.10 mm ²
Property class (S_p / S_y)	8.8 – 580 MPa / 640 MPa
Nut factor K	0.2 – Dry steel, plain
Plate 1 (head side) – Aluminum 6061-T6	8.00 mm · E 68.9 GPa · p_G 300 MPa
Plate 2 (nut side) – Mild steel (S235)	12.00 mm · E 200.0 GPa · p_G 490 MPa
Grip length $L = t_1 + t_2$	20.00 mm
External load P · applied torque T	500 N · 6.00 N·m
Washers · preload target	none (bare head) · 65% of proof

1 • Tightening — what the wrench actually does

1 • PRELOAD FROM TORQUE

$$F_i = \frac{T}{K \cdot d} = \frac{6.00 \times 10^3}{0.2 \times 6.00} = \mathbf{5,000 \text{ N}}$$

Roughly half the wrench torque is reacted in the threads and half under the head, and most of both is spent on friction — only around a tenth of it ends up as preload. K bundles all of that into one number, and it scatters $\pm 25\%$ between real joints, so read F_i as a band rather than a value.

2 • DIRECT TENSILE STRESS

$$\sigma = \frac{F_i}{A_s} = \frac{5,000}{20.10} = \mathbf{249 \text{ MPa}}$$

A_s is the tensile stress area — between the pitch and minor diameters, not the nominal $\varnothing$, because that is the section that actually carries the load through the threads.

3 • TORSION LEFT IN THE SHANK

$$\tau = \frac{16 \cdot T_{th}}{\pi \cdot d_s^3}, T_{th} \approx 0.5 \cdot T = \frac{16 \times 3.00 \times 10^3}{\pi \times 5.06^3} = \mathbf{118 \text{ MPa}}$$

Half the applied torque is reacted in the threads and twists the shank while the wrench is on. It relaxes once you let go, which is why the service check below drops it.

4 • COMBINED (REDUCED) STRESS – THE TIGHTENING CHECK

$$\sigma_{red} = \sqrt{(\sigma^2 + 3\tau^2)} = \sqrt{(249^2 + 3 \times 118^2)} = \mathbf{322 \text{ MPa}}$$

SF = $S_p / \sigma_{red} = 580 / 322 = \mathbf{1.80}$ — this is the headline number, and it is the harshest moment in the joint's life. VDI 2230 works exactly this way; Shigley reaches the same place by capping preload at 75–90% of proof instead.

2 • The two springs

Everything after tightening follows from one fact: the bolt is a stretched spring and the clamped plates are a squashed one, and they share the same joint. Their **relative** stiffness is what decides who feels an external load.

5 • BOLT STIFFNESS

$$k_b = \frac{E \cdot A_s}{L} = \frac{200.0 \times 20.10}{20.00} = \mathbf{201 \text{ kN/mm}}$$

A plain tension member over the grip length. Stretch at preload: $\Delta L = F_i / k_b = \mathbf{24.9 \text{ }\mu\text{m}}$.

6 · MEMBER STIFFNESS

$k_m = f(30^\circ \text{ cone frusta}) = 8.00 \text{ mm of Aluminum 6061-T6} + 12.00 \text{ mm of Mild steel (S235), in series} = \mathbf{576 \text{ kN/mm}}$

Shigley's pressure cone: the clamp force spreads at 30° from under the head to mid-grip and converges again on the nut, so the plates are a stack of frusta in series — each with its own modulus. Squash at preload: $\delta_m = \mathbf{8.7 \text{ }\mu\text{m}}$. The stack is **2.9x** stiffer than the bolt, and that ratio is the whole game.

7 · STIFFNESS RATIO

$$C = \frac{k_b}{k_b + k_m} = \frac{201}{201 + 576} = \mathbf{0.259}$$

The bolt's share of any external load. $C = 0.259$ means the bolt picks up **26%** of P and the plates give up the other 74%.

3 · In service — with $P = 500 \text{ N}$ on the joint

8 · LOAD SHARING

$$F_b = F_i + C \cdot P = 5,000 + 0.259 \times 500 = \mathbf{5,129 \text{ N}}$$

The bolt rises by only 129 N — this is why a preloaded joint survives fatigue: the bolt barely notices the load cycling.

9 · WHAT IS LEFT CLAMPING

$$F_m = F_i - (1 - C) \cdot P = 5,000 - 0.741 \times 500 = \mathbf{4,629 \text{ N}}$$

The external load is mostly subtracted from the clamp, not added to the bolt.

10 · SEPARATION

$$P_{\text{sep}} = \frac{F_i}{1 - C} = \frac{5,000}{0.741} = \mathbf{6,744 \text{ N}}$$

SF against separation = $P_{\text{sep}}/P = \mathbf{13.49}$. Past this load the plates lift, the bolt takes the whole of P directly, and the joint starts hammering itself apart. Keep 1.5 or more here.

11 · BEARING UNDER THE HEAD AND NUT

$$p = \frac{F}{\pi/4 \cdot (d_w^2 - d_h^2)} = \frac{5,129}{31.85} = \mathbf{161 \text{ MPa}}$$

Bare head: $d_w = 1.5d = 9.00 \text{ mm}$ over a $d_h = 1.06d$ clearance hole, so $A_{\text{bear}} = 31.85 \text{ mm}^2$. Against p_G : **1.86** on Aluminum 6061-T6, **3.04** on Mild steel (S235). A washer takes the annulus to 105.08 mm^2 — 3.3x the area for the same preload.

VERDICT AT $6.00 \text{ N}\cdot\text{M}$

CHECK	VALUE	SF
Bolt while tightening (von Mises vs proof)	322 MPa	1.80
Bolt in service (tension vs yield)	255 MPa	2.51
Joint separation	6,744 N to separate	13.49
Bearing on Aluminum 6061-T6 (head)	161 MPa	1.86
Bearing on Mild steel (S235) (nut)	161 MPa	3.04

TORQUE SPECIFICATION

PER BOLT	N · M
Bolt alone — 65% of proof (7,578 N)	9.09
Capped by bearing on Aluminum 6061-T6 (p_G 300 MPa) — preload alone, before the C·P deduction	10.32
Recommended — limited by the bolt	9.09
Same joint with washers under head and nut	9.09
You have applied	6.00

Preload & recommended torque

What proof strength is

ISO 898-1 tabulates a *proof load*: the tension a bolt carries and releases with no measurable permanent set. Proof strength is that load over the stress area. It is a tabulated column, not something derived from yield, and it sits deliberately below the 0.2% offset yield. For **8.8 (medium-carbon, Q&T)**, $S_p = 580 \text{ MPa}$ against $R_{p0.2} = 640 \text{ MPa}$ — a ratio of **0.91**, and across the classes it runs 0.88–0.91. Past proof the bolt takes a permanent set and simply loses the preload you just installed.

Why the target preload is a fraction of it

SOURCE	RECOMMENDS
Shigley — reused connections	$0.75 \cdot F_p$
Shigley — permanent connections	$0.90 \cdot F_p$
VDI 2230	$\approx 0.90 \text{ of yield} \div \alpha_A$
This calculator — your setting	$0.65 \cdot F_p$

Do not cite 0.65 as Shigley. Shigley's figure is 0.75 or 0.90, and VDI 2230 does not work in "% of proof" at all. **0.65 is this toolkit's own conservative default**, at the low end of the common 60–75% band, because these calculators are aimed at joints where a printed or light-alloy part — not the bolt — is the weak side. The selector on the model tab raises it to Shigley's figures when the joint warrants it. Two reasons the default is defensible:

Reason 1 — the headline number leaves out tightening torsion

REDUCED STRESS AT THE 65% TARGET

$\sigma_{\text{red}} = \sqrt{(\sigma^2 + 3\tau^2)} = \sigma = 0.65 \cdot 580 = 377$, plus thread torsion at $K = 0.2 = 488 \text{ MPa} = 84\% \text{ of proof}$

A "65% preload" is really **84% utilisation** while the wrench is still on. Shigley's 0.90 would put the reduced stress past proof before you let go — legitimately, because that figure assumes the torsion is accounted for elsewhere. Torsion relaxes afterwards, which is why the service check on the model tab runs on tension alone.

Reason 2 — the nut factor scatters straight onto preload

IF K LANDS...	PRELOAD / PROOF
25% low — grabbier than assumed	0.87
as assumed	0.65
25% high — slipperier	0.52

Be clear-eyed: at the unlucky end the same wrench reading installs 0.87 of proof in tension. The torsion does not grow with it — you applied the same torque — but the combined stress still reaches **102%** of proof (92% of yield), which is past the load the bolt is guaranteed to release from without a permanent set. It will not snap; it will quietly keep less preload than you think it has. Dropping the target buys margin, not certainty — that is the weakness of torque control, and it is why joints that matter use angle control past snug or measure bolt stretch.

Where the recommended torque comes from

Handbook torque tables answer "how tight?" for the *bolt*, and quietly assume the clamped parts can take it. True for steel and aluminium; false for plastics, laminates and thin castings, where the head crushes into the surface long before the bolt is in danger. So the recommendation here is the lesser of two preloads:

1 · WHAT THE BOLT WANTS — YOUR 65% TARGET

$$F_{\text{tgt}} = 0.65 \cdot S_p \cdot A_s = 0.65 \times 580 \times 20.10 = 7,578 \text{ N}$$

Fixed by the fastener and the target — M6, 8.8 at 65% — so it is the same in every row of the table below.

2 · WHAT THE CLAMPED MATERIAL ALLOWS UNDER THE HEAD

$$F_{\text{bear}} = 0.9 \cdot p_G \cdot A_{\text{bear}} = 0.9 \times 300 \times 31.85 = 8,599 \text{ N}$$

The softer of the two plates sets the cap. A_{bear} is the annulus under a bare head: $\pi/4 \cdot ((1.5d)^2 - (1.06d)^2) = 31.85 \text{ mm}^2$. The 0.9 keeps the recommendation clear of the limit it is capped by, so a torque this tool suggests is never simultaneously flagged as crushing.

2B · MINUS THE SHARE OF P THE BOLT HANDS TO THAT SAME FACE

$F_{\text{bear,net}} = F_{\text{bear}} - C \cdot P = 8,599 - 0.259 \times 500 = \mathbf{8,470\text{ N}}$

Bearing is checked in *service*, on $F_b = F_i + C \cdot P$ — the nut presses on the plate with the external load's share as well as the preload. Preload may only use what is left.

3 · THE LESSER OF THE TWO, AS A WRENCH READING

$T = K \cdot F \cdot d = 0.2 \times 7,578 \times 6.00 / 10^3 = \mathbf{9.09\text{ N}\cdot\text{m}}$

Limited by **the bolt**. Same relation, same shared fastener table and same 0.9 margin as the Cylinder Clamp calculator — plus the C·P deduction above, which only this calculator needs because only it models an external load.

Every clamped material, at your thread and grade

Run at **M6, 8.8, Dry steel, plain** — and at **your** joint's load share $C \cdot P = 129\text{ N}$, deducted from every *use* column exactly as in steps 2b-3 above, so the row for your softer plate is the model tab's recommendation — change any of those and every torque column moves. Amber means **bearing on the plate** is what holds you back, not the bolt.

CLAMPED MATERIAL	E GPA	Σ _Y MPA	P _G MPa	BEARING CAP N·M	USE, BARE N·M	WITH WASHER
Mild steel (S235)	200.0	235	490	16.85	9.09	9.09
Alloy steel (S355 / 4140)	200.0	355	760	26.14	9.09	9.09
Stainless 304 / A2	193.0	215	500	17.20	9.09	9.09
Aluminum 6061-T6	68.9	276	300	10.32	9.09	9.09
Aluminum 7075-T6	71.7	503	410	14.10	9.09	9.09
Gray cast iron (GJL-250)	110.0	165	800	27.52	9.09	9.09
Brass (CuZn37)	100.0	200	300	10.32	9.09	9.09
Ti-6Al-4V	114.0	880	900	30.96	9.09	9.09
FR-4 PCB (glass-epoxy)	12.0	300	60	2.06	1.91	6.65
POM / Delrin	3.1	70	90	3.10	2.94	9.09
ABS	2.2	45	55	1.89	1.74	6.09
ABS-PC blend	2.4	55	65	2.24	2.08	7.22
Nylon 12 (PA12)	1.7	48	50	1.72	1.56	5.52
Nylon 12 GF30 (glass-filled)	6.0	110	110	3.78	3.63	9.09
Nylon 6/6 (PA66, dry)	2.8	80	70	2.41	2.25	7.79

Reading it. *Bearing cap* is line 2 above turned into torque — how hard you could pull this bolt before the head starts sinking into that material, preload alone. *Use* is the lesser of the bolt's own target and that cap minus your load's C·P, so it flattens at **9.09 N·m** for every material where the bolt governs, and drops several-fold for polymers and laminates. That is not conservatism — it matches the separate torque tables plastics and PCB suppliers publish. The last column is the same joint with washers: 3.3× the bearing area, which is the cheapest fix there is whenever a plate flags red.

REFERENCES

Budynas & Nisbett, **Shigley's Mechanical Engineering Design**, ch. 8 – pressure-cone stiffness, load sharing, preload targets

ISO 898-1 – proof load, $R_{p0.2}$, R_m and stress area by property class

VDI 2230 Part 1 – reduced stress while tightening, tightening factor α_A , permissible surface pressure p_G

SAE J429 – inch fastener grades (proof, yield and tensile in ksi)

Typical reference values, not certified allowables. p_G for polymers is a short-term surface pressure — they creep badly above it, so it doubles as the long-term clamp limit. Verify against your own material data before production.

Design tips

1 • Tighten it properly — a loose preloaded joint is the failure

Nearly every bolted-joint failure in service is a joint that was **under-tightened**, not over-tightened. Too little preload and the plates separate under load, the bolt takes the full cycle instead of a fraction of it, and it fails in fatigue — or simply walks loose. Aim for the recommendation and treat it as a target, not a ceiling to creep up on.

Yours: recommended **9.09 N·m** (limited by the bolt) • installs **7,578 N** of preload • separation SF at your load **13.49**

2 • Stiff members, flexible bolt — that ratio is the design

$C = k_b / (k_b + k_m)$ decides how much of every external load the bolt has to carry. Low C is what you want: the load unclamps the plates instead of working the fastener. You lower it by making the **members stiffer** (steel not plastic, more material around the hole, no soft gaskets in the stack) or the **bolt longer and more slender** — which is why aerospace joints use long, thin, waisted bolts rather than short stubby ones.

Yours: $C = 0.259$ — the bolt takes **26%** of P. Double the grip to 40.00 mm and it would fall to **0.149**

3 • Washers, whenever a soft material is in the stack

The head and nut press on a small annulus, and soft plates crush there long before the bolt is in danger. A plain washer takes the bearing area from 1.5d to 2.2d across — about **3.3x the annulus** — and raises the whole torque ceiling with it. It is the cheapest fix in fastening.

Yours: bearing **161 MPa** under the head • softest plate is **Aluminum 6061-T6** (p_G 300 MPa) • SF **1.86** • the bolt already governs here, so washers buy margin against crushing rather than more torque

4 • Lubrication is a specification, not a detail

K is not a property of the bolt — it is a property of the bolt *as you assemble it*. Dry, zinc-plated, oiled and anti-seize span K ≈ 0.12 –0.22, nearly a factor of two in preload for the same wrench reading. Pick one, write it on the drawing, and use it. Lubricating a joint whose torque was specified dry over-tightens it by up to **60%**; the reverse leaves it slack.

Yours: Dry steel, plain ($K \approx 0.20$) $\rightarrow K = 0.2$. The same 6.00 N·m at $K = 0.12$ would install **8,333 N** instead of **5,000 N**

5 • Grip length: longer is better, and short joints are brittle

A short grip makes a stiff bolt (high C) and makes every bit of embedding cost a large fraction of the preload — surfaces flattening by 5.0 μm matter enormously across a 6.00 mm grip and hardly at all across 60.00 mm. Where a joint must be short, use a shoulder or a long collar to buy elastic length back.

Yours: grip **20.00 mm** = $3.3 \cdot d$ • bolt stretch at preload **24.9 μm** — losing 5.0 μm of that to embedding costs **20%** of the preload

6 • Don't over-grade the bolt

A 12.9 bolt only helps if the **bolt** is the limit. When the recommendation is capped by bearing on a plastic or a laminate, a stronger class changes nothing at all — the head still sinks into the plate at the same pressure. Spend the money on a washer, a bigger thread, or a metal insert instead. Higher classes are also more notch-sensitive and, in the case of 12.9, prone to hydrogen embrittlement when plated.

Yours: limited by **the bolt** — a higher class would raise the recommendation

7 · Threads into soft material need engagement, not torque

This calculator assumes a through-bolt with a nut, and never checks thread stripping. Tapped into aluminium you need roughly **1.5-2·d** of engagement, into plastic **2-2.5·d** or a heat-set insert, before the threads are as strong as the bolt. Below that the hole strips well under any torque on this page, and the failure is sudden and unrepairable in place.

8 · Re-torque, and never trust one reading

Surfaces embed and polymers creep: a joint checked 24 hours after assembly regularly reads 10-20% low, more in a soft or freshly-printed stack. Re-torque it. Add thread locker or a prevailing-torque nut where vibration is present — friction is what keeps a fastener from backing out, and it is the first thing a loosening cycle eats. **Where a joint failing would injure someone, torque control alone is not enough:** use angle control, load- indicating washers, or measure the stretch.

Closed-form design check, not FEA. Nut-factor torque model (K scatters $\pm 25\%$ between real joints — lubricate for consistency), fully-threaded fastener for kb, Shigley 30° cone frusta for km, concentric and purely tensile external load. Typical reference values — verify before production use.